

Surface Area of Pyramids

Lesson 10-5

Name: _____

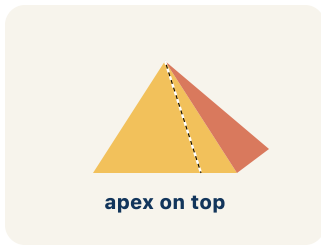
Date: _____

Class: _____

Key Vocabulary

Level 1 support

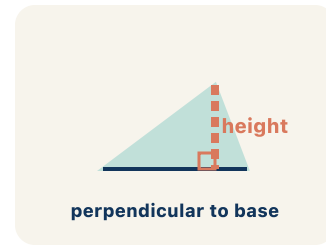
Picture first, then the word, then a plain-language meaning. Say each word out loud.



Like the Great Pyramid of Giza — a square on the bottom, 4 triangles slanting up to a point

Pyramid

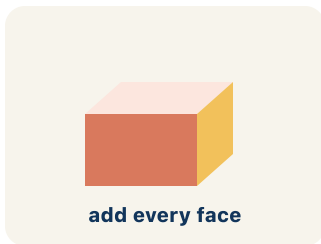
A solid with a flat bottom and triangle sides that meet at one point on top.



If you slide your finger from the bottom edge up the triangle face to the top point — that distance is the slant height

Slant height

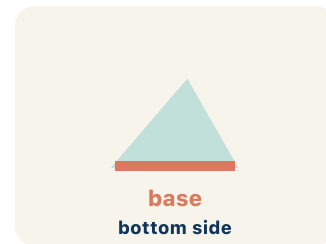
The height of a side triangle, measured along its slanted face.



A square pyramid has 4 lateral faces — one triangle for each side of the square base

Lateral face

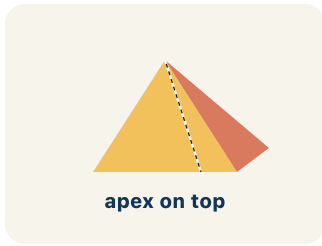
A triangle side of a pyramid, not the bottom.



A square pyramid sits on a square base; base area = side × side

Base

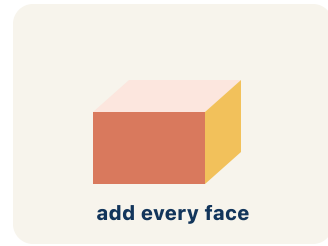
The flat bottom of a pyramid.



The pointy top of a pyramid — all the slanted edges connect here

Apex

The point at the top of a pyramid where the sides meet.



For a square pyramid: lateral area = $4 \times (\frac{1}{2} \times \text{base edge} \times \text{slant height})$

Lateral area

The total area of just the side triangles, not the bottom.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can find the surface area of a pyramid by adding the base area and the lateral faces.

1 Notice

Your class is building a pyramid-shaped display to showcase the time capsule at the school entrance.

2 Key idea

I can find the surface area of a pyramid by adding the base area and the lateral faces.

3 Words I'll use

Pyramid

Slant height

Lateral face

Base

Apex

Lateral area



Think About It

- How many faces does a square pyramid have?
- What shape is the base? What shape are the side faces?
- What is the difference between the height of the pyramid and the slant height?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: A square pyramid has a base edge of 5 in and a slant height of 7 in. What is the surface area?

A. 95 in^2

B. 70 in^2

C. 95 in^3

D. 120 in^2

Step 1

Base: $5 \times 5 = 25 \text{ in}^2$.

Step 2

Each lateral face: $\frac{1}{2} \times 5 \times 7 = 17.5 \text{ in}^2$.

Step 3

Four faces: $4 \times 17.5 = 70 \text{ in}^2$.

Step 4

SA = $25 + 70 = 95 \text{ in}^2$.



Answer: A. 95 in^2

 We do — together

Solve this one with your class using the same steps.

Problem: How many lateral (triangular) faces does a square pyramid have?

- A. 4
- B. 3
- C. 5
- D. 6

Step 1

Step 2

Step 3

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: What is the area of one triangular face with base 8 in and slant height 6 in?

- A. 24 in^2
- B. 48 in^2
- C. 14 in^2
- D. 24 in^3

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

A square pyramid time capsule display has a square base and four triangular faces that meet at a point on top. To find its surface area, which parts do you need to measure?

Level 1 support

Start here: Surface area of a pyramid is the base area plus the areas of all the triangular lateral faces.

Need a hint?

Sentence starters (optional)

I need the area of the ____ plus the areas of the ____ triangular faces.

Necesito el área de la ____ más las áreas de las ____ caras triangulares.

The surface area is the ____ area plus the ____ faces.

El área de superficie es el área de la ____ más las caras ____.

Word bank: pyramid base lateral face slant height surface area

Listen for: Students explain a square pyramid's surface area = base area + the four triangular lateral faces, and that you add every face together.

Level 2

Why do you use the slant height, not the pyramid's vertical height, to find the area of each triangular face?

- I use the slant height because it is the ____ of each triangular face.
- The vertical height would not work because ____.

EXPLORE

For Display A (square base 6×6 , slant height 8), how did you combine the base area and the four triangular faces to get the total surface area?

Level 1 support

Start here: Add the base area to four times the area of one triangular face ($\frac{1}{2} \times \text{base} \times \text{slant height}$).

 Need a hint?**Sentence starters (optional)**

The base area is $___ \times ___ = ___$, **and one triangular face is** $\frac{1}{2} \times 6 \times 8 = ___$.

El área de la base es $___ \times ___ = ___$, y una cara triangular es $\frac{1}{2} \times 6 \times 8 = ___$.

The total surface area is the base plus $4 \times ___ = ___ \text{ in}^2$.

El área total de superficie es la base más $4 \times ___ = ___ \text{ in}^2$.

Word bank: **pyramid** **base** **lateral face** **slant height** **surface area**

Listen for: Students compute base = 36 in^2 , one face = 24 in^2 , four faces = 96 in^2 , total SA = 132 in^2 , distinguishing base from lateral faces.

Level 2

Two pyramids have the same 8×8 base but different slant heights. How will their surface areas compare, and which part stays the same?

- The pyramid with the larger slant height has $___$ surface area because $___$.
- The $___$ area stays the same because slant height only affects $___$.

CONNECT

A museum builds a square pyramid display case with base edge 3 ft and slant height 4 ft, and the base is open (no glass). Talk through how to find how much glass the 4 triangular sides need.

Level 1 support

Start here: Find one triangular face area, multiply by 4, and skip the base since it is open.

💡 Need a hint?

Sentence starters (optional)

Each triangular face has area $\frac{1}{2} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ ft}^2$.

Cada cara triangular tiene área $\frac{1}{2} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ ft}^2$.

Four faces need $\underline{\hspace{1cm}} \text{ ft}^2$ **of glass, and the base needs** $\underline{\hspace{1cm}}$ **because it is open.**

Cuatro caras necesitan $\underline{\hspace{1cm}} \text{ ft}^2$ de vidrio, y la base necesita $\underline{\hspace{1cm}}$ porque está abierta.

Word bank: (pyramid) (base) (lateral face) (slant height) (surface area)

Listen for: Students compute one face = $\frac{1}{2} \times 3 \times 4 = 6 \text{ ft}^2$, four faces = 24 ft^2 , and explain the open base is not counted, so only the lateral area matters.

Level 2

How does leaving the base open change which formula you use, and how would the answer change if the base needed glass too?

- Leaving the base open means I use only the $\underline{\hspace{1cm}}$ area because $\underline{\hspace{1cm}}$.
- If the base needed glass, I would add $\underline{\hspace{1cm}}$ to the total.

REFLECT

A classmate found a square pyramid's surface area by adding only the four triangular faces. What did they forget, and how would you correct it?

Level 1 support

Start here: They left out the base area, which must be added to the lateral faces for total surface area.

 **Need a hint?**

Sentence starters (optional)

They forgot to add the ____ area to the triangular faces.

Olvidaron sumar el área de la ____ a las caras triangulares.

The correct total surface area is the base plus the ____ lateral faces.

El área de superficie total correcta es la base más las ____ caras laterales.

Word bank: pyramid base lateral face slant height surface area

Listen for: Students identify the missing base area and state that total surface area = base area + all lateral (triangular) faces.

Level 2

When is it correct to leave out the base (like an open display case), and when must it be included? Generalize a rule.

- You leave out the base when ____, but include it when ____.
- In general, total surface area of a pyramid is always ____.

Write About the Math

The Writing Revolution

I can explain my work using the words pyramid, slant height, lateral face, and base.

1. Kernel Sentence subject + verb

Model: Pyramid is a solid with a flat bottom and triangle sides that meet at one point on top.
Pirámide es un sólido con un fondo plano y lados triangulares que se unen en un punto arriba.

Write a kernel sentence about pyramid. Use a subject and a verb.

Escribe una oración base sobre pirámide. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Pyramid matters in math
Pirámide importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Pyramid matters in math because ____.
Pirámide importa en matemáticas porque ____.

but
pero

Pyramid matters in math, but ____.
Pirámide importa en matemáticas, pero ____.

so
entonces

Pyramid matters in math, so ____.
Pirámide importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about pyramid.
Di un hecho verdadero sobre pyramid.

Pyramid ____.

Question
Pregunta

Ask a question about pyramid.
Haz una pregunta sobre pyramid.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about pyramid.
Muestra entusiasmo sobre pyramid.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with pyramid.
Dile a un compañero qué hacer con pyramid.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I need the area of the ____ plus the areas of the ____ triangular faces.

Necesito el área de la ____ más las áreas de las ____ caras triangulares.

The surface area is the ____ area plus the ____ faces.

El área de superficie es el área de la ____ más las caras ____.

The base area is ____ $\times$ ____ = ____, and one triangular face is $\frac{1}{2} \times 6 \times 8 =$ ____.

El área de la base es ____ $\times$ ____ = ____, y una cara triangular es $\frac{1}{2} \times 6 \times 8 =$ ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A triangular pyramid (all faces triangles) has 4 faces each with area 9 cm^2 . What is the surface area?

- A. 36 cm^2
- B. 13 cm^2
- C. 27 cm^2
- D. 18 cm^2

Show your work:

2. To find a pyramid's surface area, you...

- A. Add the base area to the areas of all the triangular faces
- B. Multiply length by width by height
- C. Only count the base
- D. Subtract the triangles from the base

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Two pyramids have the same base area (64 in^2) but different slant heights. Pyramid A has slant height 6 in and Pyramid B has slant height 10 in. How much more surface area does Pyramid B have? Why does slant height affect surface area but NOT base area?

Sentence starter: Pyramid A: $SA = 64 + 4(\frac{1}{2} \times 8 \times 6) = 64 + \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ in}^2$. Pyramid B: $SA = 64 + 4(\frac{1}{2} \times 8 \times 10) = 64 + \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \text{ in}^2$. Pyramid B has $\underline{\hspace{1cm}}$ more in^2 because $\underline{\hspace{1cm}}$. Slant height only affects $\underline{\hspace{1cm}}$ because $\underline{\hspace{1cm}}$.

Show your work:

Reflect — Exit Ticket

A square pyramid has a base edge of 6 in and a slant height of 5 in. What is the total surface area?

- A. 96 in^2
- B. 60 in^2
- C. 96 in^3
- D. 36 in^2

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. 4 — *A square pyramid has a square base and 4 triangular lateral faces — one for each side of the square.*
2. **You Try (notes):** A. 24 in^2 — *Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 8 \times 6 = 24 \text{ in}^2$.*
3. **Try It 1:** A. 36 cm^2 — *4 faces $\times 9 = 36 \text{ cm}^2$.*
4. **Try It 2:** A. Add the base area to the areas of all the triangular faces — *Surface area = base area + the areas of all the side (triangular) faces.*
5. **Exit Ticket:** A. 96 in^2 — *Base: $6 \times 6 = 36 \text{ in}^2$. Each lateral face: $\frac{1}{2} \times 6 \times 5 = 15 \text{ in}^2$. Four faces: $4 \times 15 = 60 \text{ in}^2$. $SA = 36 + 60 = 96 \text{ in}^2$. Surface area uses square units (in^2).*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Pyramid is a solid with a flat bottom and triangle sides that meet at one point on top.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).